Singon **Kim**

AI Research Engineer

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"Cool head and warm heart."

Hi there!

I'm an AI Research Engineer focused on *Foundation Models*, *AI Agents*, and *Human-Centered Intelligence*. I'm experienced in pre-training, post-training, and data-centric AI, with a focus on building reliable, scalable systems.

I believe AI should reduce knowledge asymmetry and help people overcome their limitations. My goal is to build intelligent systems that make expertise more accessible, amplify human potential, and ultimately contribute to a happier and more empowered society.

Education

Korea University — *Seoul, S.Korea*

M.S. in Artificial Intelligence, *Advisor: Prof. Seong-Whan Lee* — *Aug. 2023 - Feb. 2026*

- Researching Context Compression for Large Language Models
- Cumulative GPA: 4.31 / 4.5

Sogang University — *Seoul, S.Korea*

B.S. in Mathematics & B.A in Economics — *Mar. 2015 - Aug. 2023*

- Cumulative GPA: 3.53 / 4.5

Work Experience

SK Telecom — *Seoul, S.Korea*

Sovereign AI Foundation Model Intern *Mentor: Seongmin Ok* — *Mar. 2026 - Jun. 2026*

- Built a synthetic data pipeline for tool-use post-training, generating high-quality and reliable trajectories.

Naver Cloud — *Seongnam, S.Korea*

Foundatoin Model Research Intern *Mentor: Hangyeol Yu* — *May 2025 - Oct. 2025*

- Developed a scalable pre-training synthetic data pipeline to generate large-scale, knowledge-rich datasets for foundation model training.
- Investigated efficient LLM architectures that achieve strong length generalization and stable performance beyond training context lengths.
- Explored distributed training frameworks and optimization techniques for large-scale foundation model training.

Projects

A.X K2 Sovereign AI Foundation Model Project — *Omnimodal Foundation Model, SKT*

National Flagship AI Development Project — *Mar. 2026 – Jun. 2026*

- Designed and implemented an execution-driven synthetic data pipeline for enhancing tool-use reasoning and agent capabilities.
- Established strict heuristic filtering and rubric-based validation frameworks to improve data reliability and mitigate hallucinations.
- Built MCP integrations utilizing K-Skill and public tools, including Korean-Law-MCP, to expand practical tool-use capabilities.

HyperCLOVA X Sovereign AI Foundation Model Project — *HyperCLOVA, Naver Cloud*

National Flagship AI Development Project — *May 2025 – Oct. 2025*

- Developed large-scale, high-quality datasets spanning diverse domains to improve foundation model knowledge coverage and generalization.
- Curated domain-specific datasets in specialized areas, including finance and nuclear power, to strengthen expert-domain capabilities.
- Designed synthetic reasoning datasets to improve mathematical reasoning and problem-solving performance.

AI Agents for CBRN Operational Command and Guidance — *PRML Lab, Korea University*

Agency for Defense Development (ADD) – National R&D Project — *Dec. 2024 – May 2025*

- Contributed to project proposal writing and early-stage project planning.
- Evaluated and deployed serving frameworks for on-premise environments.
- Developed demonstration systems to validate and showcase the initial service vision.

AI assistant for elementary teachers using Large Language Models — *PRML Lab, Korea University*

i-Scream Edu – Industry-Academia Research Project — *Sep. 2023 – Jan. 2024*

- Building High-Quality Mathematics and Science Datasets
- Construction of Short-Answer and Open-Ended Benchmarks for Elementary Education

AI Chatbot for Food Delivery Recommendations & Review generation

Naver Connect Camp AI Tech Final Project — *Nov. 2021 – Dec. 2021*

- Information retrieval from a large-scale database using Elastic Search.
- Devised a purification strategy for harmful data
- Construction of a Restaurant Metadata Dataset using Web Crawling

Research Publications

International Conference

ACoRN: Noise-Robust Abstractive Compression in Retrieval-Augmented Language Models
S. Kim, G. Jung, and SW. Lee
International Joint Conference on Neural Networks (IJCNN), 2025, 1-8 — *First Author*

arXiv: preprint

A.X K2 Technical Report
SK Telecom
arXiv: 2608.30181 — *Data Engineering*

HyperCLOVA X 32B Think
TEAM, NAVER Cloud, HyperCLOVA X
arXiv: 2601.03286. — *Internship*

HyperCLOVA X 8B Omni
TEAM, NAVER Cloud, HyperCLOVA X
arXiv: 2601.01792 — *Internship*

HyperCLOVA X THINK Technical Report
TEAM, NAVER Cloud, HyperCLOVA X
arXiv: 2506.22403 — *Contributor*

Awards

Domestic

2026 **Frontier Award (1st Place),** [Lunit] Conquer Health: Foundation Model Hackathon for Medical Science — *Seoul, South Korea*

Extracurricular Activity

Module labs - tensor2slicon — *S.Korea*

Member — *Sep. 2026 –*

- **AI Model-to-Accelerator Execution Stack**: Study and analyze the end-to-end execution pipeline from computation graphs and kernels to compilers, runtimes, and GPU/TPU/NPU architectures.
- **AI Systems Optimization Practice**: Explore memory movement, tiling, Roofline Model, and Python-based kernel DSLs through hands-on implementation and performance analysis.
- Developed open learning materials with reproducible code, diagrams, and exercises, while laying the groundwork for a future Legato user community.

Naver Connect Boost Course Study Coach — *S.Korea*

Freelance, study coach — *Jun. 2025 - Aug. 2025*

- Mentored Pre-Boost Course participants by answering technical questions related to AI, machine learning, and LLMs.
- Presented a live lecture on Transformer architectures.
- Reviewed and provided feedback on participants' LLM prompting strategies and prompt engineering assignments.

LG aimers 3rd — *S.Korea*

Member — *Jul. 2023 - Aug. 2023*

- Participated in both the online preliminary round and on-site final hackathon for online product sales forecasting.
- Studied time-series forecasting methods, including D-Linear and related forecasting architectures.

Naver Connect Boost Camp AI TECH 2nd *S.Korea*

MEMBER *Jul. 2021 – Dec. 2021*

- Studied fundamental AI concepts and methodologies required for AI engineering.
- Improved model performance through competitions in image classification, KLUE, and MRC.
- Strengthened core AI engineering skills through collaborative team-based projects.

Technical Skills

CODE LANGUAGE

python **Strong Python proficiency**, spanning AI model development (PyTorch) and backend engineering (FastAPI).

C/C++ **Intermediate proficiency**, developed through computer science coursework and systems programming.

Language

Korean **Native**,

English **Intermediate**,